DEMAK® GROUP
Resins & Dispensing Technology



CONTINUOUS VACUUM

DISPENSING EQUIPMENT FOR YOUR APPLICATIONS

Passion for Innovation

DEMAK® GROUP
Resins & Dispensing Technology



CV 55 ASI
2 Pumps



CV 170 ASI
2 Pumps

CV ASI
PATENTED TECHNOLOGY

- **CONTINUOUS VACUUM PROCESS** automatic loading/reloading of the resin from the drums for non stop production.
- **NO SETTLING**: continuous recirculation from the bottom to the top of tank with external Archimedean screw and continuous stirring of the product.

The **ASI (ANTI SETTLING INTERNAL)** system with internal pumps avoids any possible settling and allows maintenance without unloading of the tanks.

Thanks to the latest innovation process, most of DEMAK dispensing systems are designed with **CV TECHNOLOGY**.

The **CONTINUOUS VACUUM PROCESS** allows automatic loading/reloading of the resin from the drums for non stop production.

Automatic "**WORN PUMP GASKET**" alarm.

Continuous Vacuum means an **ULTRA EFFICIENT DEGASSING SYSTEM COMBINED WITH HIGH PRODUCTIVITY** for each customer.

Automatic loading of resin and hardener; automatic degassing (up to 0.2 mBar) with "**DEMAK® Continuous Vacuum System**".

DEMAK® THIN FILM DEGASSING SYSTEM with new double extra large plate to guarantee fast moisture extraction.

CV 55 - CV 170 - CV 600 huge tanks capacity to guarantee 24/7 continuous working.

DEMAK® GROUP
Resins & Dispensing Technology



CV 600 ASI
2 Pumps



Our **CV SYSTEM MODULARITY** allows you to add robots, vacuum chambers, pre-heating and polymerization ovens, as well as automatic lines to the existing dosing unit at any time.

TANK LEVEL CONTROL through load cells to avoid any maintenance of level control plunged into the resin and hardener.

CONTINUOUS VACUUM SYSTEM: 24/7 non-stop working.

NO SETTLING: continuous recirculation from the bottom to the top of tank with external Archimedean screw, continuous stirring of the product.

SINGLE MOTORIZED PUMP system that guarantees high flow pressure.



CV DL 70 ASE
up to 8 Pumps



CV 100 ASE
up to 4 Pumps

CV ASE
PATENTED TECHNOLOGY

- **SERVOMOTORS** let you control the resin flow speed in a very accurate way.
- Our product treatment system allows a **CONTINUOUS AND CONSTANT HOMOGENIZATION OF THE FILLERS**, avoiding settling anywhere in the process.

The **ASE (ANTI SETTLING EXTERNAL)** system with external pump system for fast and easy maintenance without unloading the tanks.

Automatic loading of resin and hardener; automatic degassing (up to 0.2 mBar) with "**DEMAK® Continuous Vacuum System**".

DEMAK® THIN FILM DEGASSING SYSTEM with new double extra large plate to guarantee fast moisture extraction.

CV DL 70 - CV 100 - CV 200 - CV 600 huge tanks capacity to guarantee 24/7 continuous working.

Automatic "**WORN PUMP GASKET**" alarm.

TANK LEVEL CONTROL through load cells to avoid any maintenance of level control plunged into the resin and hardener.

CONTINUOUS VACUUM SYSTEM: 24/7 non-stop working.

NO SETTLING: continuous recirculation from the bottom to the top of tank with external Archimedean screw, continuous stirring of the product.





CV 200 ASE
up to 8 Pumps

CV 600 ASE
up to 10 Pumps

- Our **CV SYSTEM MODULARITY** allows you to add at any time robots, vacuum chambers, pre-heating and polymerization ovens, as well as automatic lines to the existing dosing unit.

- **NO DOWNTIME FOR MAINTENANCE**: it involves mainly the piston inside the pumps. An automatic system (managed by the machine software) allows you to take off this part and to easily replace it.

Option of **MULTIPLE MOTORIZED PUMPS** to guarantee independent volumes.

Besides, ASE System main feature is the non-stop production with one (or more) set of pumps while performing maintenance on the others.

VARIABLE RATIO BETWEEN RESIN AND HARDENER.
Possibility of fast and precise changes of resin flow-out from nozzles.

HIGH ACCURANCY IN RESIN DISPENSING VOLUME.
Less reject.

All Continuous Vacuum machines have a **NEW GASKET SYSTEM** for the pistons with only one gasket and automatic registration during the wear-out of the same.

LOW COST MAINTENANCE (one gasket against 7 on traditional machines), longer durability thanks to the automatic registration during the working cycles.





CV 2 SMART
Stand Alone

CV 2 SMART + **VC xyz**
Vacuum Chamber

CV SMART
PATENTED TECHNOLOGY

SMALLEST UNIT
Ultra-efficient thanks to the automatic loading, degassing and dispensing.

The **CV SMART SERIES** s the newest generation of machines developed by **DEMAK®**.

The **CV SMART EQUIPMENT** performs all the main characteristics of the other series, in fact the ***CONTINUOUS VACUUM SYSTEM is always a central factor***.

The **CV2 SMART** is the smallest unit produced by **DEMAK®** and it is ultra-efficient thanks to automatic loading, degassing and dispensing. This equipment is capable to work **24/7 NON-STOP** and has a **21 lt capacity**.

The **CV2 SMART PUMPS** are driven by servomotors in order to guarantee optimal precision and the control of the dispensing speed curve, which can be regulated according to the production step and to the capability of the parts to receive resin.

The **CV2 SMART PUMPS** has been **DESIGNED ON A MODULAR CONCEPT**, so it is possible at any time to upgrade it with **XYZ ROBOT** or **VACUUM CHAMBERS** of a suitable size for the customer's application.





CV DL 70 ASI
Stand Alone

CV DL 70 ASI + **VC xyz**
Vacuum Chamber

MODULAR CONCEPT
It is possibleto upgrade it with XYZ ROBOT or VACUUM CHAMBERS at any time.

ANTI-SETTLING
Thanks to the continuous materials recirculation and single brushless motorized pump upon each tank.

One of the most complete machine of the **CV SMART** series is the **CV2 SMART + VC**: an automatic dispensing unit with a vacuum chamber provided with an integrated xyz robot.

This equipment is particularly compact and suitable for any application that needs vacuum dispensing.

The **CV DL70 ASI** has a capacity of **70 lt** in order to guarantee a higher production per hour, with all the automatic process that is the main feature of all Demak equipment.

It is possible to prepare the equipment for high viscosity resin treatment, thanks to heated jacket applied to the tank, to heated pipes and recirculation. The recirculation is very useful to avoid any material sedimentation into the pipes and all over the equipment.

CV2 SMART + VC and **CV DL70 ASI + VC** can be easily equipped with three different vacuum chambers:

- **VC 653040 xyz** : 650 x 300 x 400 mm;
- **VC 653065 xyz** : 650 x 300 x 650 mm;
- **VC 503030 xyz** : 500 x 300 x 300 mm;



PM2 CV 2 SMART
70.50.20 xyz



PM3 CV 2 SMART
140.50.20 xyz

CV SMART
PATENTED TECHNOLOGY

- **BIG FORMAT**
  From 700 x 500 to 1400 x 500 mm working area for optimal atmosphere resin casting.

The **PM2 CV2 SMART 70.50.20 xyz** is characterized by the same **CV SMART** system of the others, but it is engineered for room pressure dosing, in fact it has a ***700 x 500 x 200 mm*** working area and it is equipped with an XYZ ROBOT.

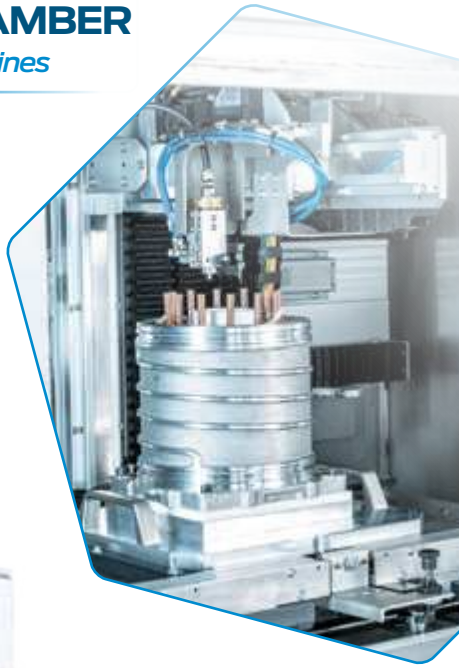
In case you need a bigger working area, we have developed the **PM3 CV2 SMART 140.50.20 xyz**, a machine equipped with a double working area than the PM2 CV2 SMART and, again, ideal for atmosphere resin casting.

ON LINE REMOTE ASSISTANCE is available on all models in order to be followed by our Technical Service at any time.

All the CV SMART series have the **CONTINUOUS VACUUM SYSTEM**, the thin film degassing system, automatic "worn pump gasket" alarm, anti-settling thanks to a continuous materials recirculation and single brushless motorized pump upon each tank.



VACUUM CHAMBER
for automatic lines





VC
VACUUM CHAMBER

VACUUM CHAMBER
stand alone

DEMAK® provides a wide range of **VACUUM CHAMBERS** with many different sizes. Movement can be performed by a single x-axis up to four axes controlled by servo-motors.

Vacuum dispensing is suggested to avoid any presence of air and moisture inside the final parts.

Possible application on an automatic line with pre-vacuum cabin, casting cabin and post-vacuum cabin.



FULL CONTROL IS NOT A OPTION!

With a **DEMAK's FULL AUTOMATIC LINE** you can obtain very high insulation grade, thanks to strong vacuum treatment of resins and vacuum potting.

DEMAK's FULL AUTOMATIC LINE gives performances, efficiency and repeatability in each step of the process.

FULLY AUTOMATIC LINES





THERMO STIRRER
60 lt

RESIN PRE-TREATMENT SYSTEM

Why it is necessary to pre-treat resin before the loading into the machine?
Because many times it has mineral fillers inside.

The **THERMO-STIRRER** unit has the purpose of heating uniformly the raw materials up to the desired temperature and to continually stir them in order to keep the fillers homogeneous and avoid any possible sedimentation at the bottom of the drums.



THERMO STIRRER
200 lt

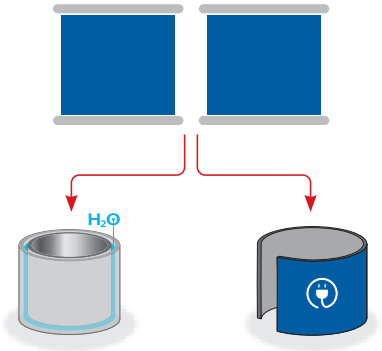
PRE-TREATMENT & HEATING UPGRADES



PLASMA
Treatment



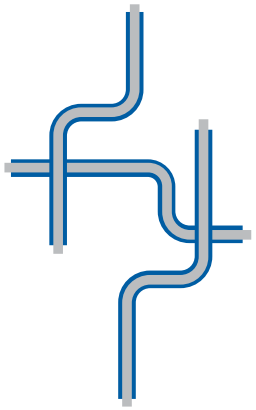
PUMPS
Heating



Water Recycling

Electric Blanket

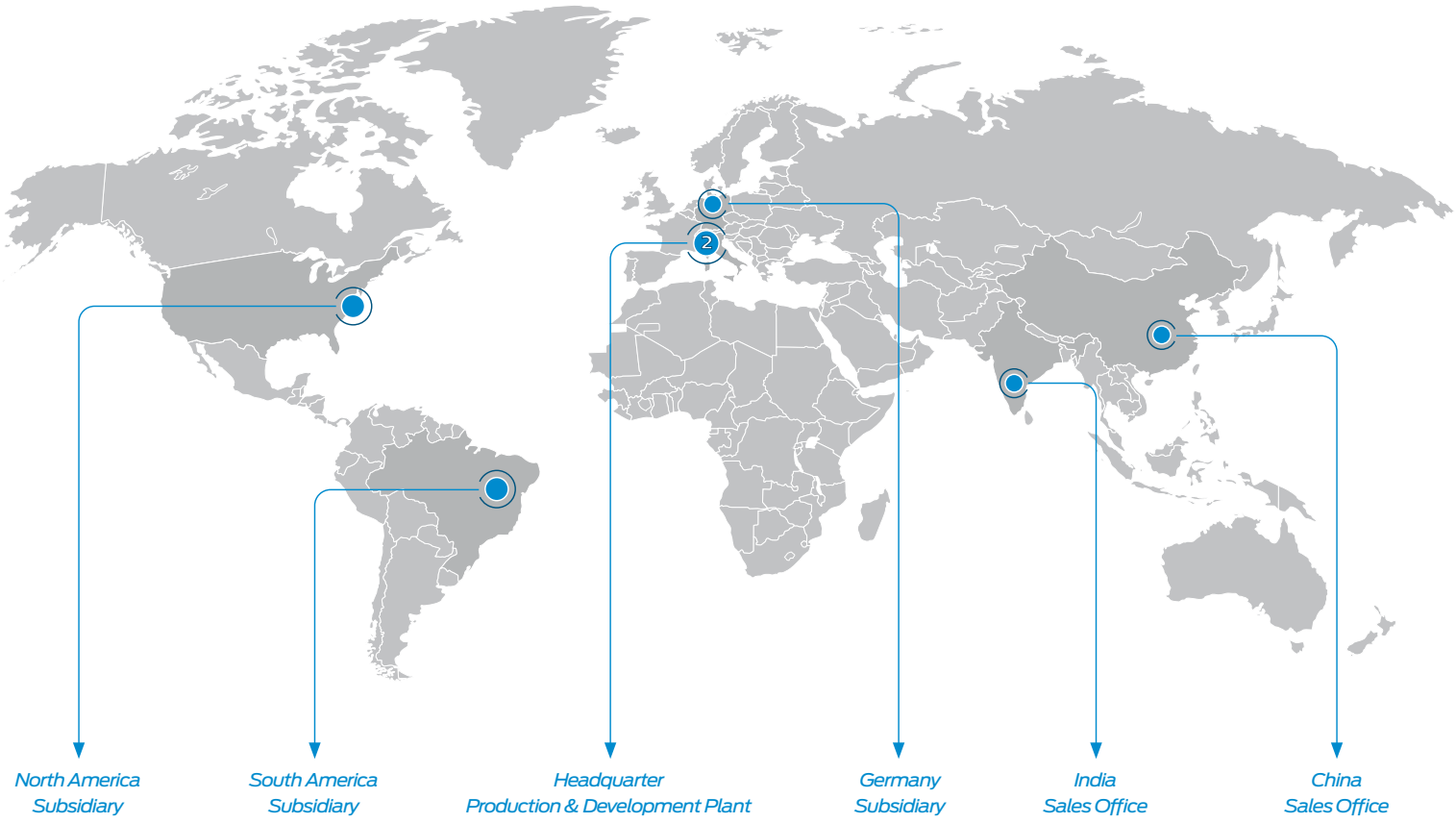
TANKS
Heating



PIPES
Heating

DEMAK® GROUP
Resins & Dispensing Technology

DEMAK GROUP – Global Presence



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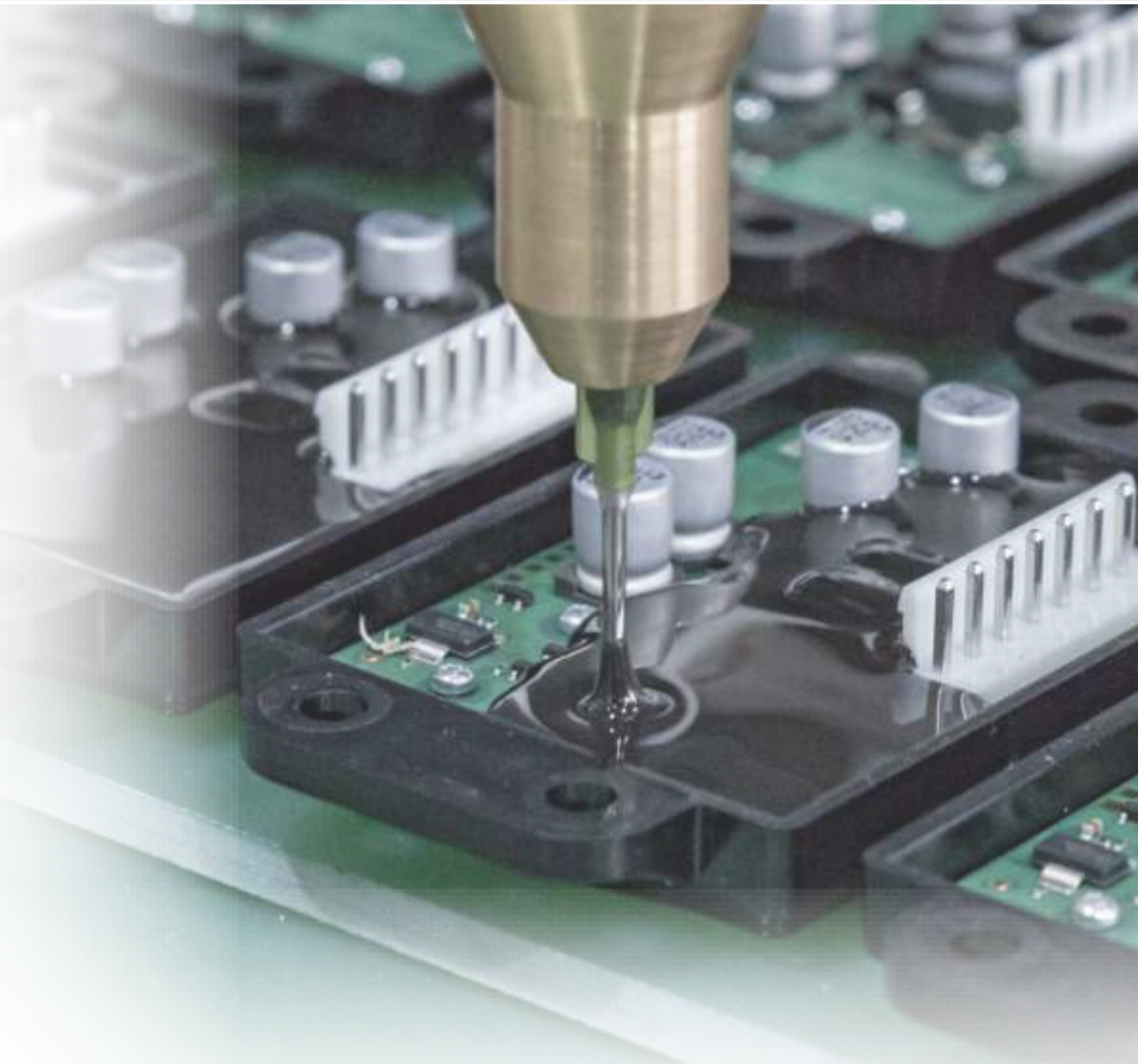
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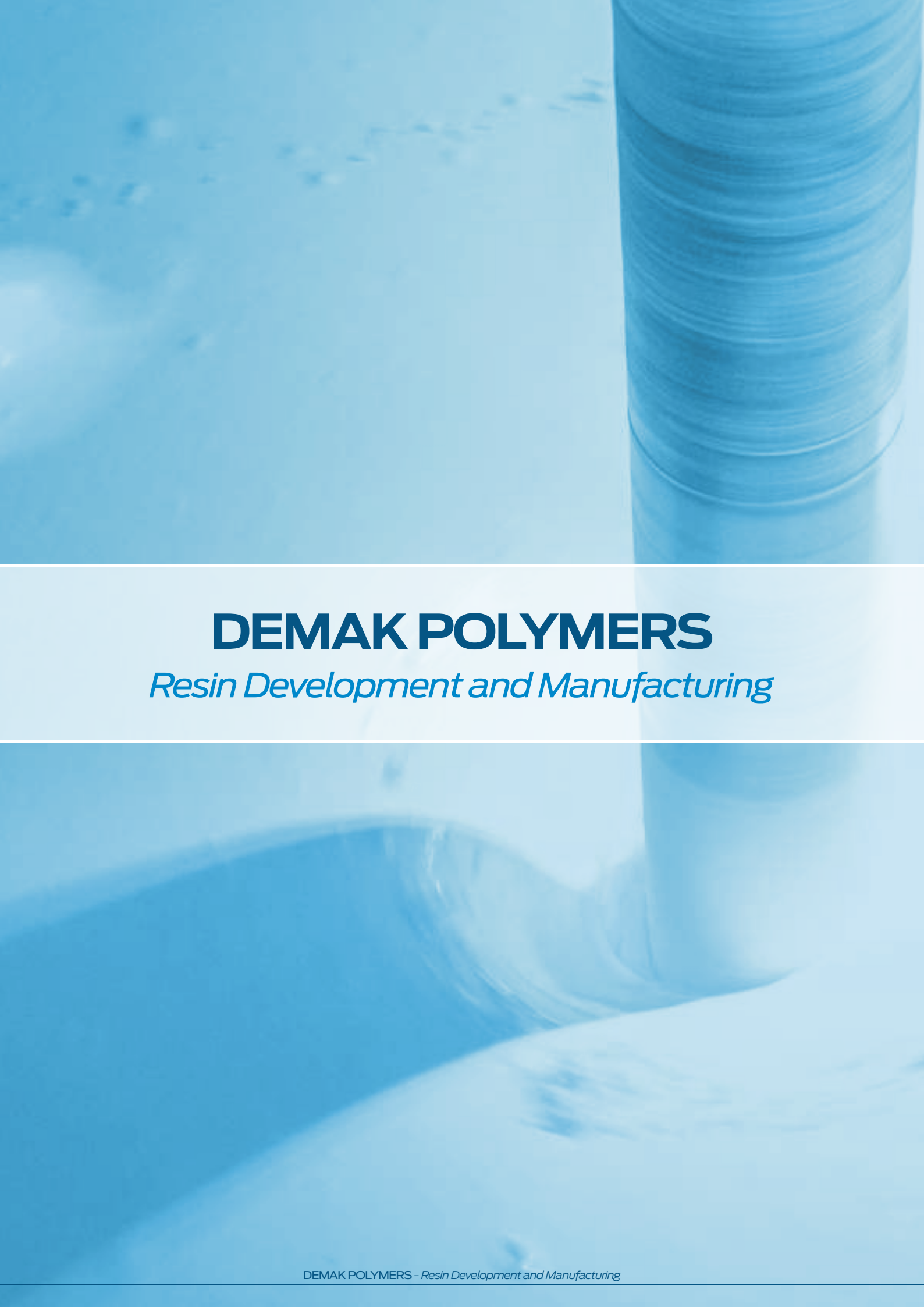
DEMAK POLYMERS

Resin Development and Manufacturing



DEMAK® GROUP

Resins & Dispensing Technology



DEMAK POLYMERS

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DEMAK POLYMERS
System-House Formulations

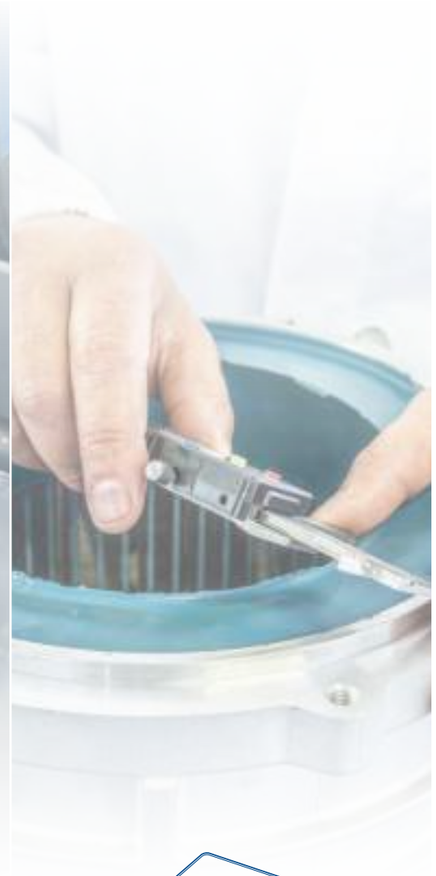
DEMAK POLYMERS, RESIN-SYSTEM-HOUSE of DEMAK GROUP since 1976, is a leading manufacturing company of advanced two-components resin systems.
Our strength is a competent team of qualified chemists dedicated to a continuous research & development and manufacturing of resin systems particularly suitable for potting technology. Quality, is a key driver we focus on, starting from the choice of the best possible raw materials to a fully automatic production area, which allows us to a complete lot traceability.
Our wide range defines Demak Polymers as the best partner in resin, able to meet even the most demanding market requirements.



R&D
LABORATORY



RESIN
MANUFACTURING



APPLICATION
ENGINEERING

DEMAK POLYMERS
Product Families



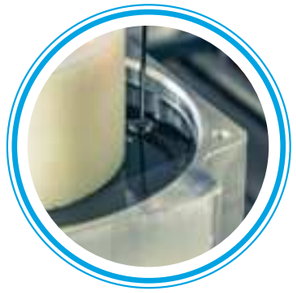


ROOM TEMPERATURE **EPOXY**

This product family has very good thermomechanical properties combined with an excellent processability which makes **DEMAPOX RT** very easy to manage and suitable for encapsulating electrical and ***electronic components***, ***motors***, ***capacitors***, ***transformers***, ***submersible pumps*** and many other applications.

HEAT CURING **EPOXY**

This product family is developed to satisfy even the most demanding requirements in terms of thermal and electrical resistance.
DEMAPOX HC is suitable for encapsulating ***electrical and electronic components***, ***EV/HEV motors***, ***capacitors***, and many other applications where a maximum impregnation of windings is necessary.



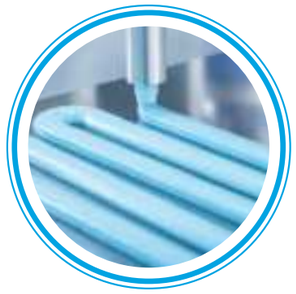


POLYURETHANE

DEMAPUR is very versatile when talking about flexibility, it maintains good thermal properties and is suitable for electrical insulation, protection, sealing and bonding. **DEMAK POLYMERS** is looking to future generations using BIO-BASED RAW MATERIALS having a lower environmental impact.

THERMAL INTERFACE **MATERIALS**

POLYURETHANE BASED GAP FILLERS with high thermal conductivity for an efficient battery thermal management.
DEMAK POLYMERS has developed several systems with different conductivity grades and performances.



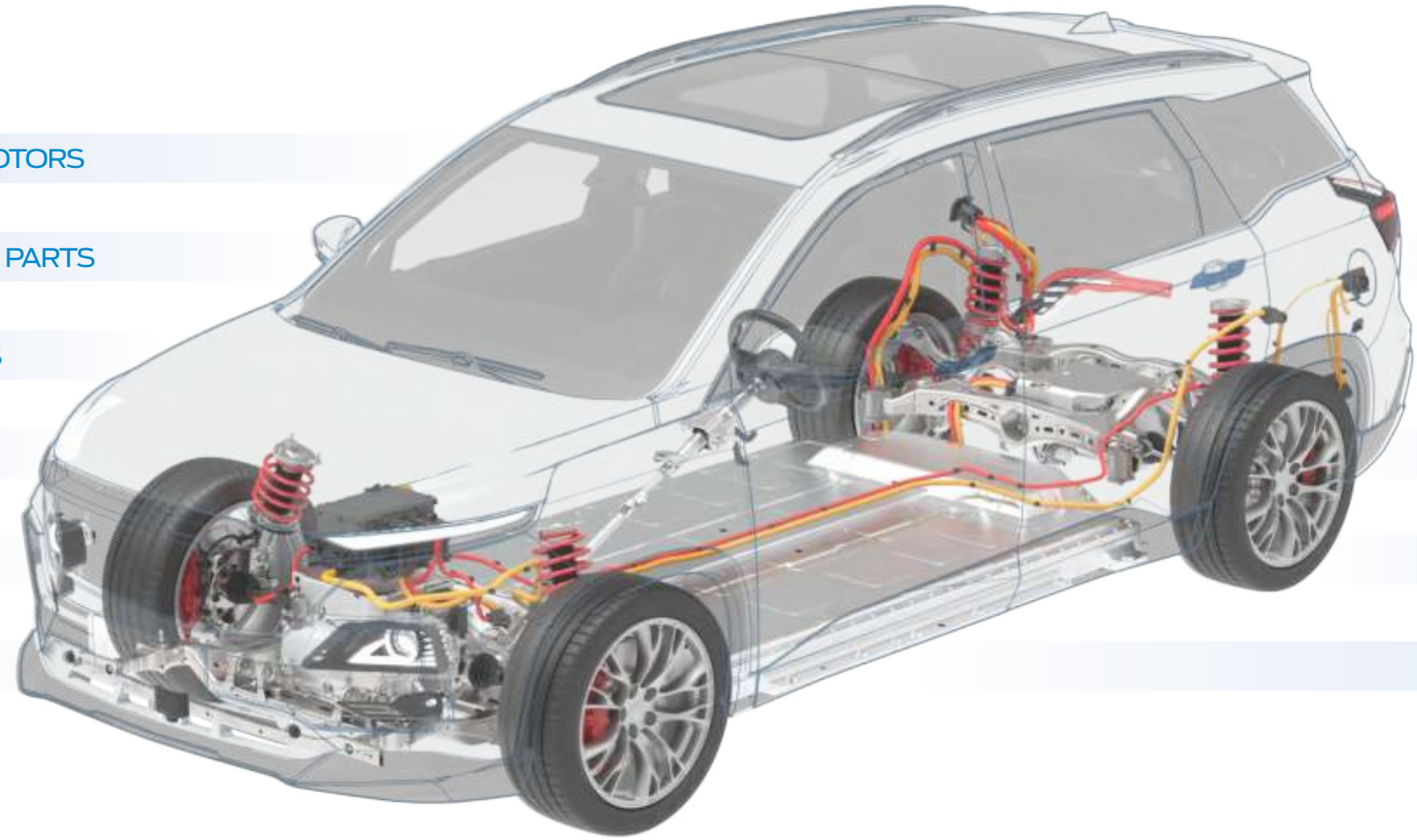
RESIN SYSTEMS
for E-MOBILITY Components

Discover more



DEMAK GROUP is working together with the main car manufacturers and their suppliers to design future electric and electronic components with potting technology.

We have combined our mechanical know-how with chemistry developing a wide range of resin systems particularly suitable for e-mobility market.



- ELECTRIC MOTORS
- ELECTRONIC PARTS
- CAPACITORS
- ECU / MCU
- SENSORS
- PCBs
- LIGHTING
- BATTERIES
- INVERTERS
- DOOR HANDLES
- TRANSFORMERS



ELECTRIC MOTORS

High performance resin systems

- Improved Heat dissipation.
- Protection from humidity and water penetration creating a sealing defence.
- Protection towards oils and lubricants or chemicals.
- Mechanical stability: harmonic vibration reduction for noise reduction.
- Keep safe from thermal shock breakage.
- Flame retardant properties if necessary.
- Improved electrical insulation.



Resin System	Chemistry	Curing	Mix Ratio	Mix Viscosity (cps)	Thermal Conductivity (W/m°K)	Glass Transition (°C)	Hardness (ShD)	UL 94 Rating	Max Operating Temperature (°C)	Description	SERVOMOTORS	INDUSTRIAL	TRACTION	ROTORS	HV
XE 206 + DK 436 NEW	EPOXY	HOT	94:100	1600 (80°C)	2	175-185	85-90	HB	200	Outstanding thermal properties, very low CTE, and excellent thermal stability. Its low viscosity ensures easy processing. Ideal for applications requiring high mechanical stability and reliable heat resistance.	•		•		•
XE 663 + DK 50/6L	EPOXY	RT	100:6.5	1500 (40°C)	1.7	70-80	85-90	HB	200	High dissipation combined with very high operating temperature when a very good flowing is needed. High cracks resistance.	•		•		•
XE 663 + DK 50/11	EPOXY	RT	100:5	1000 (40°C)	1.7	90-100	85-90	HB	200	Superior operating temperature, very high thermal conductivity. Optimal for void free encapsulation and crack resistance. High Tg.	•		•		•
XE 202 W14 + DK 433 W14	EPOXY	HOT	100:80	1300 (70°C)	1.5	110-120	85-95	V0 (file E358582)	180	High thermally conductive material, not abrasive fillers. Self extinguishing UL94 V0. Fast filling process thanks to low viscosity mixture. Electrical insulation UL file system E511310.	•	•	•		•
XE 661 HV + DK 50/11	EPOXY	RT	100:9	1000 (40°C)	1.45	95-105	85-95	V0 (file E358582)	180	High thermal resistance in particular after a post curing at 80-90°C. Very good dissipation. Electrical insulation UL file system E511310. Not abrasive fillers. Very high heat dissipation combined with low viscosity.	•	•	•		
XE 661 + DK 50/11	EPOXY	RT	100:11	750 (40°C)	1.1	90-100	80-90	V0 (file E358582)	180	High thermal resistance in particular after a post curing at 80-90°C. Very good dissipation. Electrical insulation UL file system E511310.	•	•	•		
XE 661 LV + DK 50/6L	EPOXY	RT	100:13	800 (40°C)	1.0	55-65	80-90	V0 (file E358582)	180	Very low mixture viscosity. Adapt for small parts with inner cavities.	•	•	•		
XE 2010 + DK 4320	EPOXY	HOT	100:85	1500 (70°C)	1.0	120-130	90-95	V0 (file E358582)	180	Self-extinguishing UL94 V0 (3mm) and good heat dissipation. High TG to get very good mechanical performance.		•	•		•
XE 233 VE 06 + DK 442 VE 06	EPOXY	HOT	100:80	500 (70°C)	1.0	80-90	80-90	HB	180	Outstanding low mixture viscosity that guarantees short filling process and high impregnation rate. Good thermal dissipation.	•		•		•
SEPOX 215 + DK 504	EPOXY	RT	100:13.5	1000 (40°C)	0.7	70-80	85-90	V0	155	High thermal stability at high temperatures.	•	•			•
XP 101 TC + DK 001 LV NEW	PU	RT	100:10	2000 (50°C)	1.8	20-30	65-75	V0 (file E358582)	160	Low in viscosity, high thermally conductive PU system with a wide operating temperature range from -50°C up to +160°C. UL94 V0 rated at 1.5 mm (file E358582).	•		•		•
XP 100 TC + DK 007 LV NEW	PU	RT	100:16	2500 (50°C)	1.7	15-25	60-70	V0 (file E358582)	160	CMR FREE polyurethane system, Diisocyanate concentration <0,1% according to REG (UE) 2020/1149, with very high thermal conductivity and properties.		•	•		
XP 100 HT + DK 001	PU	RT	100:18	1500 (25°C)	0.9	20-30	60-70	V0 (file E358582)	160	Polyurethane adapt for encapsulation requiring high thermal resistance and working operation in the temperature range -50°C up to 160°C. Available with no CMR Hardener. Available UL V0= 1.5 mm.		•			
FORMULATED FOR ROTORS															
XE 201 TCR + DK 432 TCH NEW	EPOXY	HOT	97:100	700 (80°C)	1.2	175-185	85-90	V0 (file E358582)	200	High Tg epoxy system with excellent mechanical stability and resistance at elevated temperatures. Designed to withstand high-speed rotation and operating temperatures, its high thermal conductivity aids efficient rotor cooling.	•		•	•	•
XE 201 TG + DK 432 TG	EPOXY	HOT	100:85	1000 (60°C)	0.8	150-160	85-95	V0 (file E358582)	200	Excellent performance at high operating temperature. Very high glass transition temperature and excellent mechanical resistance at high temperature. Epoxy with reinforcing fillers for good cracks resistance. Low viscosity mixture.	•		•	•	•
XE 201 HTG + DK 432 HTG	EPOXY	HOT	100:97	3000 (60°C)	0.8	175-185	85-95	V0 (file E358582)	200	Engineered for applications demanding high mechanical properties at elevated temperatures (i.e. ROTORS). Extremely high glass transition temperature.	•		•	•	•



CAPACITORS

High performance resin systems

- For capacitors operating in hot environment: epoxy thermal class H (180ºC).
- For capacitors testing 85ºC - 85% humidity: epoxy thermal class H (180ºC).
- For capacitors operating in low or medium temperature environment: PU.
- To avoid loss of capacitance
- Protection from humidity and water penetration creating a sealing defence.
- Protection towards oils and lubricants or chemicals and Mechanical stability.
- Keep safe from thermal shock breakage.
- Contribute to flame retardant properties if necessary.



Resin System	Chemistry	Curing	Mix Ratio	Mix Viscosity (cps)	Thermal Conductivity (W/m°K)	Glass Transition (°C)	Hardness (ShD)	UL 94 Rating	Max Operating Temperature (°C)	Description	HEV - EV	LARGE CAPACITORS	FILM CAPACITORS	POWER CAPACITORS
XE 201 EB + DK 432 EF	EPOXY	HOT	100:75	700 (70°C)	0.8	110-120	90-95	V0 (file E358582)	180	It meets the most demanding automotive requirements, offering excellent thermal shock and cycle resistance with low CTE. It combines good thermal conductivity and flow properties for fast filling and dimensional stability. Certified electrical insulation (UL file E511310).	•			
XE 201 HTG + DK 432 HTG	EPOXY	HOT	100:97	3000 (60°C)	0.8	175-185	85-95	V0 (file E358582)	200	**Engineered for applications demanding high mechanical properties at elevated temperatures**. Extremely high glass transition temperature. High thermal stability at high temperatures.	•			
XE 661 + DK 401	EPOXY	HOT	100:28	300 (70°C)	0.9	100	80-90	V0	180	Low viscosity. Fast filling. High thermal conductivity.			•	•
XE 2010 + DK 4320	EPOXY	HOT	100:85	1500 (70°C)	1.0	120-130	90-95	V0 (file E358582)	180	Self-extinguishing UL94 V0 = 3mm and good heat dissipation High TG to get very good mechanical performance.	•		•	
SEPOX 234 FR + DK 432 FR	EPOXY	HOT	100:87	1500 (70°C)	0.9	85-95	90-95	V0	180	Multi-purpose epoxy heat curing. Cost effective solution.	•		•	
SEPUR 121 FR + DK 001	PU	RT	100:33	1200 (40°C)	0.7	80-90	80-90	V0 (file E358582)	130	Hard PU with high Glass Transition temperature. Very good protection from humidity and high temperature.	•		•	•
XP 116 FR + DK 001	PU	RT	100:16	1400 (25°C)	0.65	-20°C	25-30	V0 (file E358582)	130	Non-abrasive fillers, easy to mix, excellent flowing. Soft and flexible. Self-extinguishing. Suitable for components sensitive to thermo-mechanical stress.			•	
XP 100 HT + DK 001	PU	RT	100:18	1500 (25°C)	0.9	20-30	60-70	V0 (file E358582)	160	Polyurethane adapt for encapsulation requiring high thermal resistance and working operation in the temperature range -50°C up to 160°C. Available with no CMR Hardener. Available UL V0= 1.5 mm.	•		•	•
SEPUR 114 FR + DK 001	PU	RT	100:14	1800 (25°C)	0.85	-10 / -5	35-45	V0	130	Low viscosity with **very fast setting time to get high output in production**. Elastic and therefore shock and vibration resistant. Available also in thixotropic version as sealant.			•	



ELECTRONIC COMPONENTS

High protection resin systems

- Flexibility is a must.
- Delicate components protection
- Heat dissipation.
- Protection from humidity and water penetration creating a sealing defence.
- Protection towards oils and lubricants or chemicals.
- Keep safe from thermal shock breakage.
- Flame retardant properties if necessary.
- Improved electrical insulation.



Resin System	Chemistry	Curing	Mix Ratio	Mix Viscosity (cps)	Thermal Conductivity (W/m°K)	Glass Transition (°C)	Hardness (ShD)	UL 94 Rating	Max Operating Temperature (°C)	Description	PCB	SENSORS	ECU	MODULES	DRIVERS
BD 031 FR + DK 705	PU	RT	100:9	1000 (40°C)	1.00	< -50	55 shA	V0 file E358582)	150	**Disocyanate concentration <0,1% according to REG (UE) 2020/1149**. CMR FREE flexible and elastic even at very low temperatures. The proper alternative to silicon. Adapt for application in wet conditions.	•	•		•	
BD 002 + DK 008 NEW	PU	RT	100:80	500 (25°C)	0.20	< -60	25-35 shA	HB	120	Transparent PU with very high flexibility and excellent resistance to water vapor transmission. Features very low electrical permittivity for optimal wave transparency. Variable mix ratio allows tuning to achieve maximum flexibility.	•	•		•	
SEPUR BD 020 FR + DK 001/75	PU	RT	100:18	2000 (50°C)	0.60	< -50	22-28	HB (file E358582)	130	Flexible polyurethane system maintaining flexibility at very low temperatures. It features a very stable loss factor and capacitance across a wide temperature range.	•		•		•
XP 116 FR + DK 001	PU	RT	100:16	1400 (25°C)	0.65	- 20 / -10	25-30	V0 (file E358582)	130	Multi-purpose self-extinguishing. Excellent flowability. Soft, flexible PU at very low temperature.	•	•		•	
SEPUR 114 + DK 001	PU	RT	100:14	1800 (25°C)	0.85	-10 / -5	35-45	V0	130	Low viscosity with **very fast setting time to get high output in production**. Elastic and therefore shock and vibration resistant. Available also in thixotropic version as sealant.	•			•	•
SEPUR 121 FR + DK 001	PU	RT	100:35	1000 (25°C)	0.70	75 - 85	80-90	V0 (file E358582)	130	Self-extinguishing (UL file E358582). Hard PU with high glass transition temperature. Very low viscosity. Good protection from humidity.	•				
XP 150 FR + DK 001	PU	RT	100:20	1000 (25°C)	0.80	15-20	60-70	V0 (file E358582)	130	**Self-extinguishing UL 94 V0 on 1.0 mm**. Hard PU that keeps safe from thermal shock breakage. Very low viscosity, good protection from humidity.	•			•	•
XP 102 HT + DK 001	PU	RT	100:11	3500 (25°C)	0.90	-5 / +5	25-30	V0 (file E358582)	160	**High temperature resistance and endurance: -50°C to 160°C**. Elastic PU filled with no abrasive fillers. Good thermal conductivity and **excellent chemical/ moisture resistance**. Flame retardant to UL 94 V0= 3mm. Available UL 94 V0=1.5 mm.	•		•	•	•
XP 101 TC + DK 001 LV NEW	PU	RT	100:10	2000 (50°C)	1.80	20-30	65-75	V0 (file E358582)	160	Low in viscosity, high thermally conductive PU system with a wide operating temperature range from -50°C up to +160°C. UL94 V0 rated at 1.5 mm (file E358582).	•		•		•
XP 100 HT + DK 001	PU	RT	100:18	1500 (40°C)	0.90	20-30	60-70	V0 (file E358582)	160	Polyurethane suitable for encapsulation requiring **high thermal resistance** and working operation in a temperature range beween -50°C and 160°C. Tough and stable. Available with no CMR hardener. Available UL 94 V0=1.5 mm.	•		•	•	•
XP 106 + DK 007 NEW	PU	RT	100:12	1200 (50°C)	1.00	-5/+5	15-25	V0 (file E358582)	160	CMR FREE Diisocyanate concentration <0,1% according to REG (UE) 2020/1149. Elastic PU filled with no abrasive fillers. Good thermal conductivity and excellent chemical and moisture resistance. Flame retardant properties: UL 94 V0= 3mm. Available UL 94 V0=1.5 mm	•		•		•
XE 661 + DK 50/11	EPOXY	RT	100:11	750 (40°C)	1.25	90-100	80-90	V0 (file E358582)	180	High thermal resistance RT epoxy, especially after post-curing at 80-90 °C. Very good dissipation. Electrical insulation UL file system E511310.		•	•		
XE 442 LV + DK 602	EPOXY	RT	100:20	1500 (25°C)	1.00	-5 / +5	45-55	V0	155	Very low mixture viscosity. Flexible system, soft and stable over time. No abrasive fillers. Very good thermal shock resistance.	•			•	
SEPOX 216 + DK 601	EPOXY	RT	100:30	500 (50°C)	0.50	12-15	25-30	HB	155	**Outstanding low viscosity, low density**. Epoxy able to avoid stress on circuit due to the **very high flexibility and low shrinkage**.	•			•	
SEPOX 233 FR BK + DK 601	EPOXY	RT	100:13	2500 (50°C)	1.10	10-20	50-60	V0	155	Widely used in the automotive field. Good thermal conductivity. Medium hard epoxy system, medium low viscosity.	•		•	•	
SEPOX 2100 + DK 527	EPOXY	RT	100:30	1300 (25°C)	0.55	45 - 50	75-85	HB	130	No shrinkage during polymerization. Designed to provide excellent electrical and mechanical properties, good wetting and impregnation. Resistant to humidity, chemicals and hot water.		•			

UV LINE 1K NEW

											PCB	SMD
SECRI UV 020	PU Acrylic	UV LED	1K	450 (25°C)	0.25	-10	40	VOC FREE	(-40) +120	PU-acrylic system with low viscosity and fast processability. It offers good adhesion, reliable electrical properties, and operates from -40 °C to +120 °C. Ideal for applications requiring quick curing and ease of use.	•	
SECRI UV 001	PU Acrylic	UV LED	1K	1500 (25°)	0.30	5	20	VOC FREE	(-40) +120	Low-viscosity transparent encapsulant with low shrinkage and strong adhesion. Offers high moisture resistance, good electrical properties, and is completely VOC-free. Ideal for sensitive electronic applications.	•	
SECRI UV 002	PU Acrylic	UV LED	1K	12000 (25°C)	0.30	-30	85 shA	VOC FREE	(-40) +120	Very fast-curing encapsulating resin (cures in seconds) designed for high film build over 5 mm on SMD components. Offers high transparency, excellent adhesion on various substrates, and a very low dielectric constant.	•	•
SECRI UV 002 DAM	PU Acrylic	UV LED	1K	THIXOTROPIC (25°C)	0.30	-25	85 shA	VOC FREE	(-40) +120	Thixotropic version of UV002 for high film build applications, featuring anti-sagging properties. Its high viscosity makes it ideal for creating dams (DAM) to simplify the encapsulation of components.	•	•



BATTERIES

Protection & Performance

- For an improved Thermal Management.
- Gap Filling or Adhesive function with good thermal conductivity.
- Flexibility in encapsulants to protect in vibrations and bumping.
- Protection from humidity and water penetration creating a sealing defence.
- Flame retardant properties.



Type	Resin System	Chemistry	Curing	Mix Ratio	Mix Viscosity (cps)	Thermal Conductivity (W/m°K)	Glass Transition (°C)	Hardness (ShD)	UL 94 Rating	Max Operating Temperature (°C)	Description
THERMAL INTERFACE MATERIAL GAP FILLERS	GP 300 + DK GP 725 NEW	PU	RT	100:50 pbv	450000 (25°C)	3.0	<-40	50 shA	HB	130	GAP FILLER with high thermal conductivity (3.0 W/mK) UL94 V0 in very thin layer. CMR FREE GAP FILLER.Diisocyanate concentration <0,1% according to REG (UE) 2020/1149.
	GP 2002 + DK 004 NEW	PU	RT	100:10 pbv	400000 (25°C)	2.3	-30	50 shA	V0	130	GP 2002 + DK 004 is a very soft, fast-reacting polyurethane based gap filler with high thermal conductivity and Shore A 50 hardness. Formulated without abrasive fillers for easy dispensing, it cures at room temperature and is UL94 V-0 rated, with a max operating temperature of 130 °C.
	GP 200 + DK GP 720	PU	RT	100:50	300.000 (25°C)	2.2	< -20	75 shA	V0	130	**Diisocyanate concentration <0,1% according to REG (UE) 2020/1149**. CMR FREE GAP FILLER. High thermal dissipation. UL94V0 in very thin layer.
	XP 115/S01 + DK706	PU	RT	100:6	150.000 (25°C)	1.85	-10	20-30	V0	130	**Diisocyanate concentration <0,1% according to REG (UE) 2020/1149**. **Thixotropic PU** with low specific gravity. Very high thermal conductivity. UL94 V0=1 mm. CMR FREE.
	XP 115 FR THIXO + DK 001	PU	RT	100:7	60.000 (25°C)	1.7	-10	40	V0	130	**Thixotropic PU** with low specific gravity, high heat dissipation. UL94 V0=1mm.
	SEPUR 114 THIXO + DK 001	PU	RT	100:14	1800 (25°C)	0.85	-10 / -5	35-45	V0	130	Low viscosity thixotropic gap filler/adhesive with very fast setting time to get high output in production. Elastic and therfore shock and vibration resistant. Good adhesion to several supports including metal.
ADHESIVES	MF 285 J + DK 925 NEW	PU	RT	50:100 pbv	1300 (25°C)	0,25	50-55	65-75	HB	120	Structural adhesive with outstanding elongation at break and excellent adhesion on a wide range of substrates. Offers very good electrical properties and is CMR-free, with diisocyanate content <0.1% per REG (EU) 2020/1149.
	MF 285 J + DK 925 THIXO NEW	PU	RT	50:100 pbv	THIXOTROPIC	0,25	50-55	65-75	HB	120	Thixotropic version of MF285J with excellent anti-sagging properties, ideal for vertical or high-build applications. CMR FREE gap filler with diisocyanate content <0.1% in compliance with REG (EU) 2020/1149.
	XE 442 LV + DK 602	EPOXY	RT	100:20	1500 (25°C)	1,0	-5 / +5	45-55	V0	155	**Very low mixture viscosity. Flexible system, soft and stable over time. No abrasive fillers. Very good thermal shock resistance.**
RESIN POTTING MATERIALS	BD 031 FR W + DK 705	PU	RT	100:9	1500 (40°C)	0.6	< -50	10-15	V0 (file E358582)	150	**Flexible PU with outstanding flame resistance and low specific gravity. Diisocyanate concentration <0,1% according to REG (UE) 2020/1149**. CMR FREE flexible and elastic even at very low temperature.
	BD 031 FR + DK 705	PU	RT	100:9	1000 (40°C)	1.0	< -50	55 shA	V0 (file E358582)	150	**Diisocyanate concentration <0,1% according to REG (UE) 2020/1149**. CMR FREE. The proper alternative to silicon. Adapt for application in wet conditions.
	XP 116 FR + DK 001	PU	RT	100:16	1400 (25°C)	0.65	-20 / -10	25-30	V0 (file E358582)	130	Multi-purpose self-extinguishing. Excellent flowability. Soft, flexible PU at very low temperature.
	XP 132 FR LIGHT + DK 705	PU	RT	100:34	3500 (40°C)	0.20	< -35	10-15	V0	130	**Lightweight PU with CONSTANT VOLUME** (0.55 kg/l for batteries weight production). **Diisocyanate concentration <0,1% according to REG (UE) 2020/1149**. CMR FREE.
	XP 102 RW + DK 001	PU	RT	100:8	1500 (40°C)	0.95	-5 / +5	45 shA	V0 (file E358582)	160	**RE-ENTERABLE SPECIFICALLY DEVELOPED to be easly removed to access the component and modifiy or replace it.** High temperature resistance and endurance: -50°C to 160°C. Elastic PU filled with no abrasive fillers. Flame retardant to UL 94 V0= 1.6 mm
	XP 100 HT + DK 001	PU	RT	100:18	1500 (40°C)	0.9	20-30	60-70	V0 (file E358582)	160	Polyurethane suitable for encapsulation requiring **high thermal resistance** and working operation in a temperature range beween -50°C and 160°C. Tough and stable. Available UL 94 V0=1.6 mm
	XP 100 TC + DK 007 LV NEW	PU	RT	100:16	2500 (50°C)	1.7	15-25	60-70	V0 (file E358582)	160	CMR FREE polyurethane system,Diisocyanate concentration <0,1% according to REG (UE) 2020/1149, with very high thermal conductivity and properties.
	XP 150 FR + DK 001	PU	RT	100:20	1000 (25°C)	0.8	15-20	60-70	V0 (file E358582)	130	**Self-extinguishing UL 94 V0 on 1.0 mm**. Hard PU that keeps safe from thermal shock breakage. Very low viscosity, good protection from moisture.



INDUCTIVE COMPONENTS

Improved protection

- Better mechanical stability.
- Protection from humidity and water penetration creating a sealing defence.
- Protection towards oils and lubricants or chemicals.
- Flame retardant properties if necessary.
- Improved electrical insulation.



Resin System	Chemistry	Curing	Mix Ratio	Mix Viscosity (cps)	Thermal Conductivity (W/m°K)	Glass Transition (°C)	Hardness (ShD)	UL 94 Rating	Max Operating Temperature (°C)	Description	TRANSFORMERS	FILTERS	CHOCKES	FERRITE INDUCTORS
XE 442 LV + DK 602	EPOXY	RT	100:20	1500 (25°C)	1.0	-5 /+5	45-55	V0 (file E358582)	155	Very low mixture viscosity. Flexible system, soft and stable over time. No abrasive fillers. Very good thermal shock resistance. Bisphenol free version.		•	•	•
XE 442 HV + DK 602	EPOXY	RT	100:11	2500 (40°C)	1.65	-5 /+5	55-60	V0 (file E358582)	155	**Very high thermal conductivity, provides good heat dissipation.** No abrasive fillers. Very good thermal shock resistance. Bisphenol free version.		•	•	•
SEPOX 233 FR BK + DK 601	EPOXY	RT	100:13	2500 (50°C)	1.1	10-20	50-60	V0	155	Widely used in the automotive field. Very good thermal conductivity. Medium hard epoxy system, medium low viscosity.		•	•	•
SEPOX 233 FR + DK 442	EPOXY	HOT	100:30	700 (60°C)	1.0	35-45	60-70	V0 (file E358582)	180	Flexible epoxy with extremely high thermal performance, compliant with the most demanding **railway standards EN 45545 HL3 R22/R23**. Features a variable mix ratio to enhance flexibility and adhesion.	•			•
XE 661 HV + DK 50/6L	EPOXY	RT	100:9	1100 (40°C)	1.45	55-65	80-90	V0 (file E358582)	180	Not abrasive fillers. Very high heat dissipation combined with low viscosity. Slow reactivity for big volumes. Electrical insulation UL file system E511310.	•	•	•	
SEPOX 213 FR + DK 503	EPOXY	RT	100:13	1200 (40°C)	0.9	50-60	80-90	V0 (file E358582)	155	Multi-purpose epoxy, room temperature curing system. Low density. Electrical insulation UL file system E511310.	•	•	•	
XP 116 FR + DK 001	PU	RT	100:16	1400 (25°C)	0.65	-20 / -10	25-30	V0 (file E358582)	130	Multi-purpose PU, low viscosity to get excellent flowability. Soft, flexible at very low temperature. Low density. Good protection from humidity.	•			•
SEPUR 114 FR + DK 001	PU	RT	100:14	1800 (25°C)	0.7	-10 / -5	35-45	V0	130	Low viscosity with **very fast setting time to get high output in production**. Elastic and therefore shock and vibration resistant. Available also in thixotropic version as sealant.	•			•
XP 102 HT + DK 001	PU	RT	100:11	3500 (25°C)	0.9	- 5 / + 5	25-30	V0 (file E358582)	160	**High thermal resistance** and endurance: -50°C to 160°C. Elastic PU filled with no abrasive fillers. Good thermal conductivity and good adhesion. Excellent chemical and moisture resistance. Flame retardant to UL94 V0=3mm. Available UL94 V0=1.5 mm.		•	•	•
XP 100 TC + DK 007 LV NEW	PU	RT	100:16	2500 (50°C)	1,7	15-25	60-70	V0 (file E358582)	160	CMR-FREE polyurethane system, Diisocyanate concentration <0,1% according to REG (UE) 2020/1149, with very high thermal conductivity and properties.		•	•	•
XP 101 TC + DK 001 LV NEW	PU	RT	100:10	2000 (50°C)	1.8	20-30	65-75	V0 (file E358582)	160	Low in viscosity, high thermally conductive PU system with a wide operating temperature range from -50°C up to +160°C. UL94 V0 rated at 1.5 mm (file E358582).		•	•	•
XP 100 HT + DK 001	PU	RT	100:13	1500 (40°C)	0.9	20-30	40-50	V0 (file E358582)	160	**High thermal resistance** and endurance: -50°C to 160°C. Hard PU filled with no abrasive filers. Good thermal conductivity and **excellent chemical and moisture resistance**. Good flowability. Flame retardant to UL94 V0=3mm. Available UL94 V0=1.5 mm).		•	•	•
XP 150 FR + DK 001	PU	RT	100:20	1000 (25°C)	0.8	15-20	60-70	V0 (file E358582)	130	**Self-extinguishing UL 94 V0 = 1.0 mm**. Hard PU that keeps safe from thermal shock breakage. Very low viscosity, good protection from humidity.				•
BD 031 FR + DK 705	PU	RT	100:9	1000 (40°C)	1.0	< -50	55 shA	V0 (file E358582)	150	**Diisocyanate concentration <0,1% according to REG (UE) 2020/1149**. CMR FREE flexible and elastic even at very low temperature. The proper alternative to silicon. Adapt for application in wet conditions.		•	•	•

WE ARE CERTIFIED TO MAKE YOU STRONGER

In compliance with worldwide norms and directives.

DEMAK POLYMERS provides continuous technical and processes assistance.
Being our resin in compliance with latest worldwide Norms and Directives concerning chemical products, we supply our resins with certifications and documents required by the final user as well as updates of legislations.

The QUALITY SYSTEM of DEMAK POLYMERS is certified according to:

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QUALITY SYSTEM CERTIFICATION DNV IATF 16949

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DEMAK POLYMERS has always paid attention to sustainability together with innovation and we are very attentive to improve our way of manufacturing low environmental impact polyurethanes for example, using bio-based raw materials.

DEMAK POLYMERS LOOKING AHEAD

BIO-BASED RESIN SYSTEMS

Use of bio-based raw material for lower environmental impact

CUSTOMER CARE

Constant Quality and Reliability.



Beside checking the chemical-physical parameters of the preparations, in our laboratories we also carry out further testing: simulation of environmental conditions, electrical tests, physical-mechanical analysis, application trials as well as specific tests on demand thus providing an additional service for a full customer satisfaction.





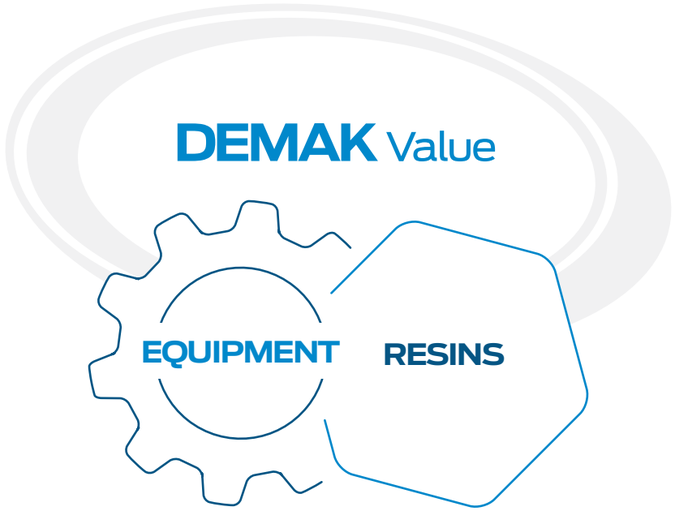


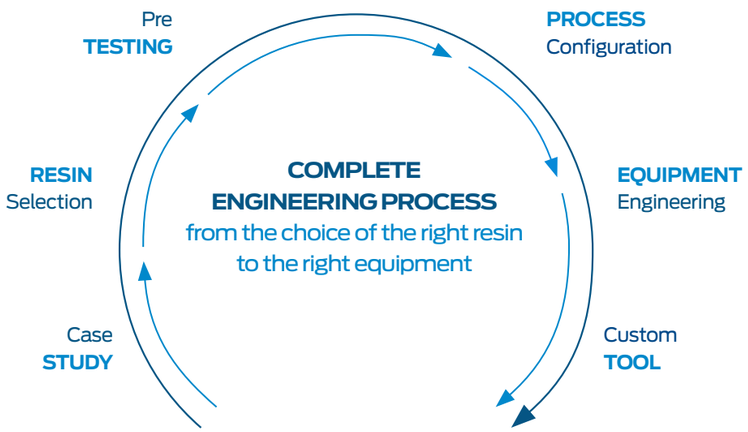
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Chemicals & Mechanical expertise at your service.

Relying on a supplier which is resins and machines manufacturer in an integrated manner, you will be able to rely entirely on a single partner who can both mechanically and chemically advise you, without having to independently study the properties of each resin in relation to your own products.
Furthermore, an integrated approach is one of the most important factors to allow the process to be carried out as efficiently as possible, minimizing waste and increasing quality.

An efficient vacuum dispensing process, together with a resin system having high thermal resistance, thermal conductivity, thermal shock resistance and many other properties allows qualifying electric components for all durability tests required by OEMs.





POTTING SPRINT

Our unique consulting approach.



DEMAK POTTING SPRINT is the unique consultancy approach that allows our top experts to provide you with the best potting process configuration for all e-mobility components.

This means a complete process industrialization, from the choice of the most suitable resin system to the correct equipment selection, always starting from your specific needs.



PIECE DESIGN CONSULTING

DEMAK approaches projects with an accurate analysis of the customer piece production requirements.



RESIN SYSTEMS

DEMAK offers a wide range of Polyurethanes, Epoxy and Heat curing Epoxy able to fulfil all kind of potting needs.



DISPENSING EQUIPMENT & TOOLING

DEMAK manufactures resin dispensing equipment from compact stand-alone machines up to fully automatic lines.

DO YOU NEED SUPPORT IN POTTING PROCESS?

Ask DEMAK.

DEMAK combined know-how, mechanical and chemical, is the key to set up a complete potting process choosing one-only reliable partner.

DEMAK manufactures machines suitable for any productivity needs. From compact machines to fully automatic lines, all our dispensing machines are equipped with the heart of Demak technology: **CONTINUOUS VACUUM**.
Resins are always properly treated in order to get the best possible impregnation to protect your components and enhance performance.

We own the **CHEMISTRY**, we own the **PROCESS**, we own the **SOLUTION**.





COMPACT Solutions



MID Production



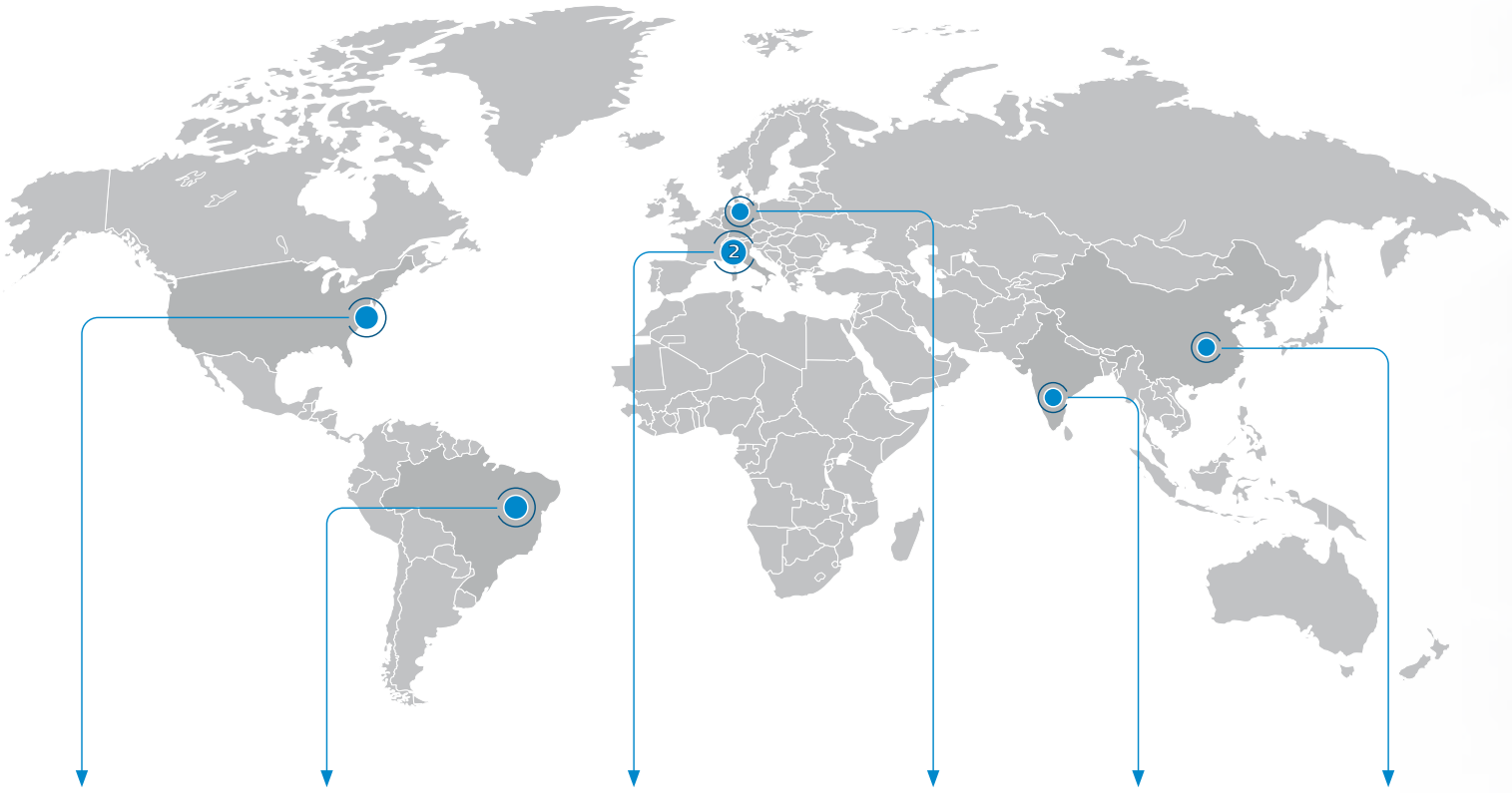
MASS Production

DEMAK POLYMERS

Resin Development and Manufacturing

DEMAK GROUP

Global Presence



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